Darshan UNIVERSITY

Python for Data Science - 2305CS303

Lab - 3

Roll No. : 122

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1. WAP to print 1 to 10.

In [1]:
```
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

2. WAP to print 1 to n.

In [3]:
```
n=int(input("enter n: "))
for i in range(1,n+1):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

3.WAP to print odd numbers between 1 to n.

In [2]:
```
n=int(input("enter n: "))
for i in range(1,n+1,2):
    print(i)
```

```
1
3
5
7
9
```

4. WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

In [4]:
```
n1=int(input("enter n: "))
n2=int(input("enter n2: "))
for i in range(n1,n2+1):
    if i%2==0 and i%3!=0:
        print(i)
```

```
10
14
16
20
```

5. WAP to print sum of 1 to n numbers.

In [8]:
```
n1=int(input("enter n: "))
sum=0
for i in range(1,n1+1):
    sum+=i
print(sum)
```

```
55
```

6.WAP to print sum of series 1 + 4 + 9 + 16 + 25 + 36 + ...n.

In [10]:
```
n1=int(input("enter n: "))
sum=0
for i in range(1,n1+1):
    sum+=i*i
print(sum)
```

```
5
```

7. WAP to print sum of series 1 – 2 + 3 – 4 + 5 – 6 + 7 ... n.

In [11]:
```
n1=int(input("enter n: "))
sum=0
for i in range(1,n1+1):
    if i%2==0:
        sum-=i
    else:
        sum+=i
print(sum)
```

```
4
```

8. WAP to print multiplication table of given number.

In [12]:
```
n1=int(input("enter n: "))
for i in range(1,11):
    print(f"{n1} * {i}= {n1*i}")
```

```
10 * 1= 10
10 * 2= 20
10 * 3= 30
10 * 4= 40
10 * 5= 50
10 * 6= 60
10 * 7= 70
10 * 8= 80
10 * 9= 90
10 * 10= 100
```

9. WAP to find factorial of the given number.

In [13]:
```
n1=int(input("enter n: "))
sum=1
for i in range(n1,1,-1):
    sum*=i
print(sum)
```

```
120
```

10. WAP to find factors of the given number.

In [14]:
```
n1=int(input("enter n: "))
for i in range(1,n1+1):
    if n1%i==0:
        print(i)
```

```
1
2
5
10
```

11. WAP to find whether the given number is prime or not.

In [16]:
```
n1=int(input("enter n: "))
# flag=True
for i in range(2,n1):
    if n1%i==0:
        print("not prime number")
        break
else:
    print("prime number")
```

```
prime number
```

12. WAP to print sum of digits of given number.

In [18]:
```
n1=int(input("enter n: "))
sum1=0
```

```
while n1>0:
    digit=n1%10
    sum1+=digit
    n1/=10
print(int(sum1))
```

```
6
```

13. WAP to check whether the given number is palindrome or not.

In [37]:
```
n1=input("enter n: ")
n2=int(n1[::-1])
n1=int(n1)
if n1==n2:
    print("palindrome")
```

```
palindrome
```

14. Fibonacci Number

In [64]:
```
n1=int(input("enter n: "))
a, b = 0, 1
for i in range(n1):
    print(a)
    a, b = b, a + b
```

```
0
1
1
2
3
```

15. Armstrong Number

In [73]:
```
n1=int(input("enter n: "))
l=len(str(n1))
sum=0
t=n1
while n1>0:
    div=n1%10
    sum+= div**l
    n1//=10
if sum == t:
    print("it is Armstrong num")
else:
    print("not armstrong num")
```

```
it is Armstrong num
```

In [ ]: